

AWS Cloud Practical

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1. Deploy a static website on two EC2 instances in different Availability Zones and configure an Application Load Balancer (ALB) to distribute traffic between them to achieve high availability.

- Launch two EC2 instances named Instance A and Instance B in the same VPC but in different Availability Zones (AZs) using public subnets. Allow inbound HTTP traffic on port 80.
- Different AZs are used to achieve high availability. If one AZ fails, traffic can still be served from the other AZ.

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EC2 > Instances > i-009c6424c1e5a7e33

Instance summary for i-009c6424c1e5a7e33 (Instance A) [Info](#)

[Refresh](#) [Connect](#) [Instance state](#) [Actions](#)

Updated less than a minute ago

Instance ID i-009c6424c1e5a7e33	Public IPv4 address 13.61.194.204 open address	Private IPv4 addresses 10.0.13.114
IPv6 address -	Instance state Running	Public DNS ec2-13-61-194-204.eu-north-1.compute.amazonaws.com open address
Hostname type IP name: ip-10-0-13-114.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-13-114.eu-north-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 13.61.194.204 [Public IP]	VPC ID vpc-024f8931266570790 (MyVPC)	Auto Scaling Group name -
IAM role -	Subnet ID subnet-02c17cda1d166a85b (Public-Subnet)	

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EC2 > Instances > i-0d64b8843ede94518

Instance summary for i-0d64b8843ede94518 (Instance B) [Info](#)

[Connect](#) [Instance state](#) [Actions](#)

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Instance ID i-0d64b8843ede94518	Public IPv4 address 16.170.166.26 open address	Private IPv4 addresses 10.0.20.195
IPv6 address -	Instance state Running	Public DNS ec2-16-170-166-26.eu-north-1.compute.amazonaws.com open address
Hostname type IP name: ip-10-0-20-195.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-20-195.eu-north-1.compute.internal	
Answer private resource DNS name -	Instance type t3.micro	Elastic IP addresses -
Auto-assigned IP address 16.170.166.26 [Public IP]	VPC ID vpc-024f8931266570790 (MyVPC)	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
IAM role -	Subnet ID subnet-061485a8046389ab0 (Public-Subnet-2)	Auto Scaling Group name -

- Install the Apache web server (httpd) on both EC2 instances, then start and enable the service.

```
[ec2-user@ip-10-0-13-114 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository                249 kB/s | 31 kB    00:00
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-13-114 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:00:20 ago on Sat Feb 14 19:17:10 2026.
Dependencies resolved.

=====
Package                        Architecture  Version                               Repository  Size
=====
Installing:
httpd                          x86_64       2.4.66-1.amzn2023.0.1                amazonlinux 47 k
Installing dependencies:
apr                            x86_64       1.7.5-1.amzn2023.0.4                amazonlinux 129 k
apr-util                       x86_64       1.6.3-1.amzn2023.0.2                amazonlinux 97 k
apr-util-ldap                  x86_64       1.6.3-1.amzn2023.0.2                amazonlinux 13 k
=====
```

```
[ec2-user@ip-10-0-13-114 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-13-114 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-13-114 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-02-14 19:17:47 UTC; 27s ago
     Docs: man:httpd.service(8)
  Main PID: 25837 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 1067)
   Memory: 13.3M
      CPU: 77ms
    CGroup: /system.slice/httpd.service
            └─25837 /usr/sbin/httpd -DFOREGROUND
              └─25838 /usr/sbin/httpd -DFOREGROUND
                └─25839 /usr/sbin/httpd -DFOREGROUND
                  └─25840 /usr/sbin/httpd -DFOREGROUND
                    └─25841 /usr/sbin/httpd -DFOREGROUND

Feb 14 19:17:47 ip-10-0-13-114.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server.
Feb 14 19:17:47 ip-10-0-13-114.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Feb 14 19:17:47 ip-10-0-13-114.eu-north-1.compute.internal httpd[25837]: Server configured, listening on: port 80
[ec2-user@ip-10-0-13-114 ~]$
```

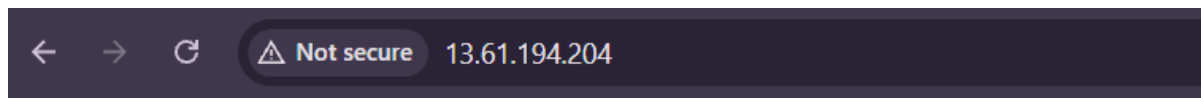
- Create a webpage named index.html inside the /var/www/html directory on both instances.
Add the content:

- “Hello from Instance A” on Instance A
- “Hello from Instance B” on Instance B

```
GNU nano 8.3 /var/www/html/index.html
<h1>Hello from Instance A</h1>
```

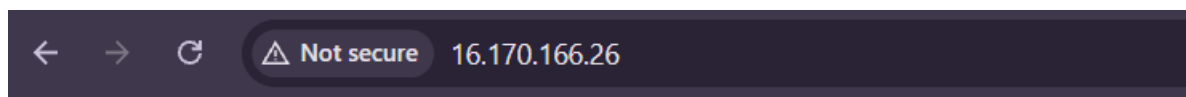
```
GNU nano 8.3 /var/www/html/index.html
<h1>Hello from Instance B</h1>
```

- Verify that both webpages are accessible using the public IP addresses of the respective instances.
 - Webpage of Instance A



Hello from Instance A

- Webpage of Instance B



Hello from Instance B

- Create a Target Group in the same VPC as the EC2 instances. Set the Health Check path to / and register both Instance A and Instance B as targets.

EC2 > Target groups > Create target group

Step 1

Create target group

Step 2 - recommended

Register targets

Step 3

Review and create

Review and create

Review your target group configuration before creating

Step 1: Target group details

Target group details

Name	Target-Group-01	Target type	Instance
Protocol : Port	HTTP: 80	Protocol version	HTTP1
VPC	vpc-024f8931266570790	IP address type	IPv4

Health check details

Health check protocol	HTTP	Health check path	/
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Edit

EC2 > Target groups > Create target group

Health check protocol HTTP	Health check path /
Health check port traffic-port	Interval 30 seconds
Timeout 5 seconds	Healthy threshold 5
Unhealthy threshold 2	Success codes 200

Step 2: Register targets [Edit](#)

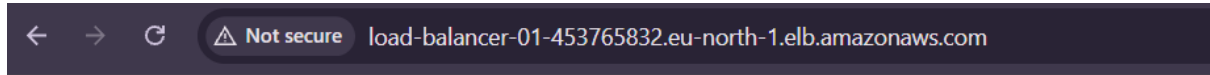
Targets (2)

Instance ID	Name	Port	Zone
i-0d64b8843ede94518	Instance B	80	eu-north-1c
i-009c6424c1e5a7e33	Instance A	80	eu-north-1a

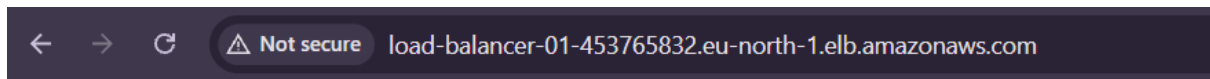
[Cancel](#)
[Previous](#)
[Create target group](#)

- Create an Application Load Balancer (ALB) as Internet-facing in the same VPC. Select two subnets that are located in different Availability Zones.
- Attach a Security Group to the ALB that allows HTTP (port 80) from 0.0.0.0/0. Configure the Listener on HTTP port 80 and forward traffic to the created Target Group.

- Access the application using the Load Balancer's DNS name.
Refresh the page multiple times to observe traffic being distributed between Instance A and Instance B.



Hello from Instance B



Hello from Instance A